PHARMACOLOGY · MCQ PRACTICE

# Laxatives and Anti-Diarrhoeal Agents

LG 3
Dr. Nermin Jamal
GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

**HOW TO USE THIS SET**

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

**4 Easy · 10 Medium · 6 Hard** · suggested time ≈ 25 minutes.

## Q Questions

**1. A patient asks what the practical definition of constipation is. What does the lecture state?** EASY

**A.** A reduced frequency or ease of stool passage relative to what is normal or expected for that individual
**B.** A complete and total absence of any bowel movements for more than one full week
**C.** Passage of hard stools accompanied by visible blood in the stool
**D.** Fewer than one bowel movement occurring per calendar month
**E.** Any single bowel movement requiring more than 5 minutes to complete

**2. A patient with mild constipation is advised on first-line non-drug management. Which of the following is recommended?** EASY

**A.** Immediate initiation of a stimulant laxative without trying non-drug measures first
**B.** A fibre-rich diet, increased hydration, minimizing tea/coffee, and physical exercise
**C.** Strict bed rest in order to reduce overall bowel activity
**D.** A high-caffeine diet specifically intended to stimulate peristalsis
**E.** Complete fluid restriction until symptoms resolve on their own

**3. A patient is prescribed senna for constipation. Approximately how long after oral intake does bowel evacuation typically occur?** EASY

**A.** 15 to 30 minutes
**B.** 1 to 2 hours
**C.** 8 to 10 hours
**D.** 24 to 48 hours
**E.** 3 to 5 days

**4. A patient asks which drug is the first-line choice specifically for traveler's diarrhoea. What is the correct answer?** EASY

**A.** Loperamide
**B.** Diphenoxylate + atropine
**C.** Activated charcoal
**D.** Bismuth subsalicylate
**E.** Methylcellulose

**5. A patient on long-term extended-release medication is started on a laxative for chronic constipation. What specific pharmacokinetic risk does the lecture highlight regarding laxative use?** MEDIUM

**A.** Laxatives substantially increase first-pass hepatic metabolism of all co-administered oral drugs, regardless of formulation or dosing schedule
**B.** Laxatives inhibit the gastrointestinal absorption of all orally administered drugs uniformly, without exception
**C.** Laxatives substantially increase renal clearance of all co-administered drugs via tubular secretion
**D.** Laxatives have no measurable effect whatsoever on the pharmacokinetics of any other co-administered drug
**E.** Accelerated intestinal transit increases loss of pharmacologic effect for poorly absorbed, delayed-acting, or extended-release oral formulations

**6. A patient in an acute surgical abdomen is mistakenly given a laxative. What is the major risk described in the lecture?** MEDIUM

**A.** Fatal intestinal rupture
**B.** Mild, self-limiting electrolyte imbalance
**C.** Transient hypotension only
**D.** Temporary loss of appetite
**E.** Mild skin rash

**7. A patient asks how bisacodyl works as a laxative. What is its mechanism?** MEDIUM

**A.** It forms a hydrophilic gel in the large intestine that distends the bowel wall
**B.** It is a potent colon stimulant acting directly on nerve fibers in the colonic mucosa to induce motility
**C.** It is a nonabsorbable salt that draws fluid into the bowel lumen via osmosis
**D.** It is a surface-active surfactant that emulsifies directly with the stool
**E.** It directly lubricates the intestinal wall to ease the passage of stool

**8. A patient taking lactulose for constipation asks how it works. What is the correct mechanism?** MEDIUM

**A.** It directly and selectively stimulates nerve fibers located within the colonic mucosa itself, inducing motility
**B.** It is absorbed systemically into the bloodstream and acts centrally in the brainstem to increase motility
**C.** Colonic bacteria metabolize it to lactic, formic, and acetic acids, increasing osmotic pressure and causing fluid accumulation and soft stools
**D.** It forms a hydrophilic gel in the large bowel that distends the intestinal wall and reflexively stimulates peristalsis
**E.** It acts as a surface-active surfactant that emulsifies directly with the stool, softening it for easier passage

**9. A patient is prescribed docusate sodium for constipation prevention. What should they be told about its onset of action and primary use?** MEDIUM

**A.** It acts within about 1 hour and is best reserved for acute, urgent constipation relief
**B.** It acts immediately upon rectal administration, within just a few minutes
**C.** It is generally considered contraindicated for any prophylactic use whatsoever
**D.** It takes days to become effective and is used primarily for prophylaxis rather than acute treatment
**E.** It only works effectively when combined together with a stimulant laxative

**10. A patient is prescribed lubiprostone for chronic constipation. What distinguishes this drug from other laxative classes regarding long-term use?** MEDIUM

**A.** It must be used only for short-term treatment due to high dependency risk
**B.** It is contraindicated in chronic constipation
**C.** It causes significant systemic electrolyte imbalance
**D.** It requires combination with a stimulant laxative for efficacy
**E.** It carries no risk of tolerance or dependency

**11. A patient asks why polyethylene glycol (PEG) solutions are used before colonoscopy. What is their primary clinical application?** MEDIUM

**A.** As electrolyte solutions used for colonic lavage to clear the gut prior to radiological or endoscopic procedures
**B.** As a stimulant laxative that directly induces colon muscle contraction and increases motility overall
**C.** As a stool softener intended for long-term chronic prophylactic use at home
**D.** As an antidiarrheal agent specifically used to reduce intestinal motility
**E.** As a lubricant solution that coats and protects the intestinal wall directly

**12. A patient with chronic diarrhoea asks how diphenoxylate and loperamide reduce bowel movements. What is their shared mechanism?** MEDIUM

**A.** They adsorb toxins and bacteria directly within the lumen of the intestine itself
**B.** They stimulate presynaptic opioid receptors in the enteric nervous system, inhibiting acetylcholine release and decreasing smooth muscle peristalsis
**C.** They inhibit prostaglandin synthesis directly within the intestinal mucosa lining, reducing secretion and inflammation
**D.** They form a steep osmotic gradient that actively draws water out of the stool
**E.** They coat and protect the intestinal mucosa directly against irritation and damage

**13. A patient asks why loperamide, unlike other opioid analogs, does not cause systemic analgesic or CNS effects at typical clinical doses. What explains this?** MEDIUM

**A.** It is rapidly destroyed by gastric acid before systemic absorption
**B.** It is metabolized entirely by colonic bacteria before absorption
**C.** It acts peripherally and does not cross the blood-brain barrier
**D.** It binds only to peripheral serotonin receptors, not opioid receptors
**E.** It has no activity at opioid receptors at all

**14. A patient asks which three core approaches the lecture describes for managing severe acute diarrhoea. What are they?** MEDIUM

**A.** Surgical resection of the affected bowel segment, broad-spectrum antibiotics, and total parenteral nutritional support
**B.** Strict fasting for several days, complete bed rest, and scheduled opioid analgesia
**C.** Laxative withdrawal, strict dietary fibre restriction, and routine antiemetic therapy
**D.** Fluid balance maintenance (ORS), anti-infectives for infectious etiology, and spasmolytics for motility reduction
**E.** Colonic lavage before endoscopy, bulk laxatives, and prophylactic stool softeners

**15. A patient taking docusate sodium asks how it produces softer stools. What is its precise mechanism?** HARD

**A.** It is metabolized by colonic bacteria into osmotically active organic acids that draw in water
**B.** It forms hydrophilic gels in the large intestine that cause water retention and reflexive bowel distension
**C.** It is hydrolyzed in the small intestine into an irritant compound that increases peristalsis
**D.** It activates luminal chloride channels directly to drive fluid secretion into the lumen
**E.** It acts as a surface-active surfactant that emulsifies with stool, producing softer feces for easier passage

**16. A patient asks why lubiprostone has minimal drug interactions despite its systemic chloride-channel activity. What property explains this?** HARD

**A.** It is rapidly metabolized in the stomach and jejunum before systemic distribution
**B.** It is not absorbed from the gastrointestinal tract at all under normal conditions
**C.** It binds irreversibly to circulating plasma proteins, which prevents drug interactions
**D.** It is excreted entirely unchanged by the kidneys within just a few minutes of dosing
**E.** It has no pharmacological activity on any receptor located outside the colon itself

**17. A patient is prescribed bismuth subsalicylate and later notices dark discoloration of the tongue and stools. What is the cause of this finding, and is it concerning?** HARD

**A.** Gastrointestinal bleeding — this requires urgent evaluation
**B.** Bismuth sulfide formation — this is a harmless side effect
**C.** Hepatotoxicity — this requires immediate drug discontinuation
**D.** Melanin deposition from a drug hypersensitivity reaction
**E.** Iron accumulation from chronic overdose

**18. A patient asks why bismuth subsalicylate, despite being effective for traveler's diarrhoea, is generally regarded as weaker than antimotility agents, and what additional precaution applies to its use?** HARD

**A.** It is actually more effective than antimotility agents but cannot be used alongside any other medication
**B.** It is roughly equally as effective as antimotility agents but carries a notable risk of toxic megacolon
**C.** It is significantly less effective than antimotility agents and can adsorb and interfere with the absorption of other co-administered oral drugs
**D.** It is less effective than antimotility agents but has essentially no interactions with other co-administered drugs
**E.** It is considered contraindicated specifically in traveler's diarrhoea due to its comparatively poor efficacy there

**19. A patient asks why activated charcoal and methylcellulose are useful as adsorbents in diarrhoea management, beyond simply binding toxins. What additional property does methylcellulose have in this context?** HARD

**A.** It also stimulates colonic peristalsis directly
**B.** It also activates luminal chloride channels
**C.** It also inhibits presynaptic opioid receptors
**D.** It also absorbs excess water and increases stool bulk
**E.** It also inhibits prostaglandin synthesis in the mucosa

**20. A patient on diphenoxylate + atropine (Lomotil) asks why atropine is included in the combination given its antidiarrheal irrelevance. What is the rationale discussed for combining it with diphenoxylate?** HARD

**A.** It directly enhances the antidiarrheal effect of diphenoxylate at the gut wall
**B.** It prevents toxic megacolon by blocking muscarinic receptors in the colon
**C.** It is required to activate diphenoxylate into its active metabolite
**D.** It prevents aspiration pneumonia during oral administration
**E.** It is combined with diphenoxylate to discourage abuse of the opioid component

## ✓ Answer key

| Q | Answer | Difficulty | Why |
|---|---|---|---|
| 1 | **A** | EASY | Constipation is practically defined as a reduced frequency or ease of stool passage relative to what is regarded as normal or expected for that individual. |
| 2 | **B** | EASY | First-line non-pharmacological management includes a fibre-rich diet, increased fluid intake, minimizing tea/coffee, and physical exercise (which activates abdominal muscles and intestinal peristalsis). |
| 3 | **C** | EASY | Senna, a widely used stimulant laxative, causes bowel evacuation within 8 to 10 hours of oral intake. |
| 4 | **D** | EASY | Bismuth subsalicylate (Pepto-Bismol) is the first-line choice for traveler's diarrhoea. |
| 5 | **E** | MEDIUM | Accelerated intestinal transit from laxatives increases the loss of pharmacologic effect for poorly absorbed, delayed-acting, or extended-release oral formulations, since they may pass through before being absorbed. |
| 6 | **A** | MEDIUM | The lecture warns that laxatives in an acute surgical abdomen can lead to fatal intestinal rupture; laxatives are contraindicated in organic/mechanical GIT obstruction. |
| 7 | **B** | MEDIUM | Bisacodyl (suppositories or enteric-coated tablets) is a potent colon stimulant acting directly on nerve fibers in the colonic mucosa to induce motility. |
| 8 | **C** | MEDIUM | Lactulose is a semisynthetic disaccharide that resists GI enzymes; colonic bacteria metabolize it to lactic, formic, and acetic acids, increasing osmotic pressure and causing fluid accumulation, colon distension, soft stools, and defecation. |
| 9 | **D** | MEDIUM | Docusate (a stool softener/emollient) takes days to become effective and is used primarily for prophylaxis rather than acute treatment. |
| 10 | **E** | MEDIUM | Lubiprostone, a chloride channel activator, is approved for chronic constipation and carries no risk of tolerance or dependency, unlike many other laxative classes. |
| 11 | **A** | MEDIUM | Polyethylene glycol (PEG) solutions are electrolyte solutions used as colonic lavage to clear the gut prior to radiological or endoscopic procedures. |
| 12 | **B** | MEDIUM | Diphenoxylate and loperamide are opioid analogs that stimulate presynaptic opioid receptors in the enteric nervous system, inhibiting acetylcholine release and significantly decreasing smooth muscle peristalsis. |
| 13 | **C** | MEDIUM | Loperamide acts peripherally and does not cross the blood-brain barrier, explaining its lack of systemic analgesic effects at typical clinical doses, unlike centrally-acting opioids. |
| 14 | **D** | MEDIUM | The three core approaches in severe acute diarrhoea are: maintaining fluid and electrolyte equilibrium (ORS), specific anti-infective agents for infectious etiology, and spasmolytics for motility reduction and symptomatic relief. |
| 15 | **E** | HARD | Docusate sodium and docusate calcium are surface-active surfactants that emulsify with stool, producing softer feces for easier passage — distinct from osmotic, bulk-forming, irritant, or chloride-channel mechanisms. |
| 16 | **A** | HARD | Lubiprostone is rapidly metabolized in the stomach and jejunum before systemic distribution, which is noted as contributing to its minimal drug interactions. |
| 17 | **B** | HARD | Bismuth subsalicylate causes harmless darkening of the tongue and black stools due to bismuth sulfide formation — not a sign of bleeding or toxicity. |
| 18 | **C** | HARD | Bismuth subsalicylate is significantly less effective than antimotility agents, and can adsorb and interfere with the absorption of other co-administered oral drugs, in addition to its mucosal coating and mild antimicrobial action. |
| 19 | **D** | HARD | Methylcellulose, used as an adsorbent in diarrhoea management, also absorbs excess water and increases stool bulk, in addition to adsorbing toxins, bacteria, and other harmful substances. |
| 20 | **E** | HARD | Diphenoxylate is often combined with atropine to discourage abuse of the opioid component — atropine's unpleasant anticholinergic effects at supratherapeutic doses deter misuse. |